Predicting Heart Failure Mortality: A Comparative Analysis of Machine Learning and Deep Learning Models

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Abstract

Predicting heart failure mortality is critical for clinical decision-making. This study compares a Multi-Layer Perceptron (MLP) with traditional Logistic Regression. Using a clinical dataset (n=299), we found that Logistic Regression achieved a superior accuracy of 81.7%, outperforming the MLP model (75.0%).

1 Methodology

1.1 Data Preprocessing

To ensure reproducibility, we implemented a median imputation for missing data and utilized **Stratified Sampling** (80/20 split). Features were standardized using Z-score scaling to facilitate the convergence of the MLP model.

1.2 Exploratory Analysis and Workflow

Before modeling, we analyzed feature correlations to understand underlying clinical relationships. Figure 1 displays the correlation heatmap of the clinical records. The overall experimental methodology follows the pipeline illustrated in Figure 2.

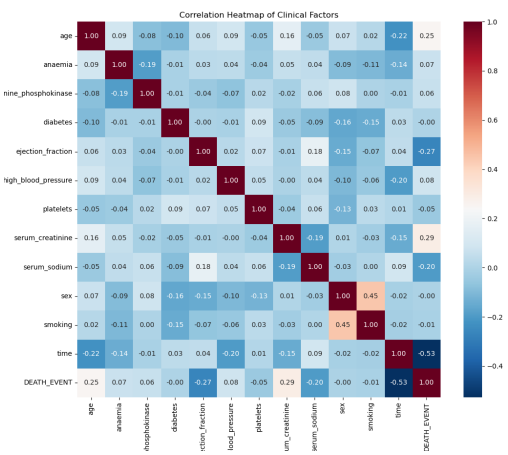

Figure 1: Correlation heatmap of clinical features.

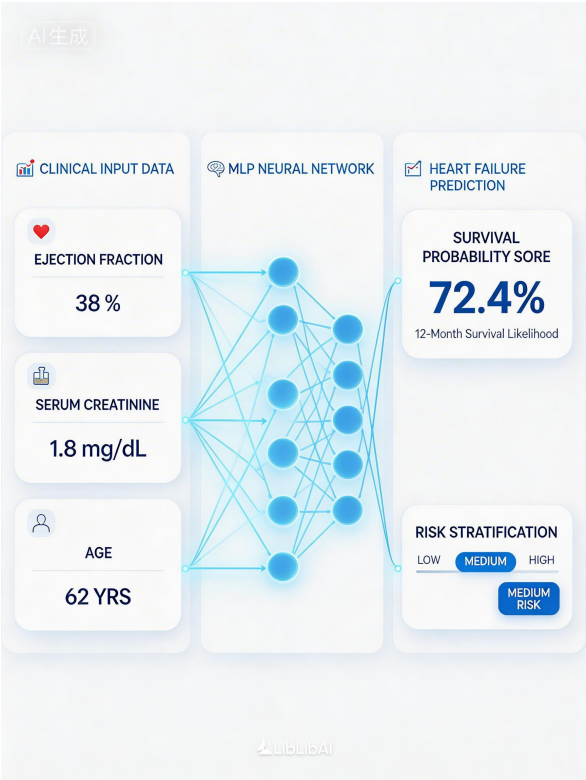

Figure 2: Proposed AI workflow for heart failure prediction.

2 Results

2.1 Feature Importance

As shown in Figure 3, Random Forest analysis identified **Serum Creatinine** and **Ejection Fraction** as the most critical predictors of mortality.

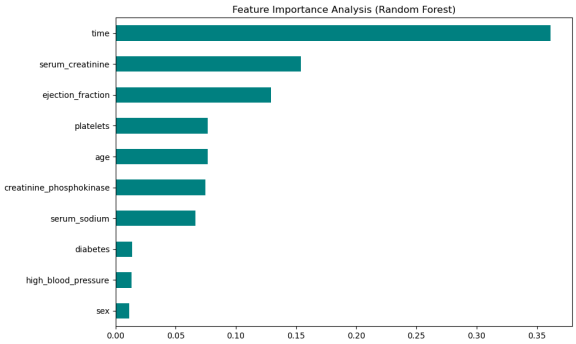

Figure 3: Clinical features ranked by Random Forest importance.

2.2 Performance Evaluation

Table 1 details the comparative metrics. The results indicate that traditional models remain robust for small-scale clinical datasets. To further illustrate the models' discriminative ability, Figure 4 presents the Receiver Operating Characteristic (ROC) curves.

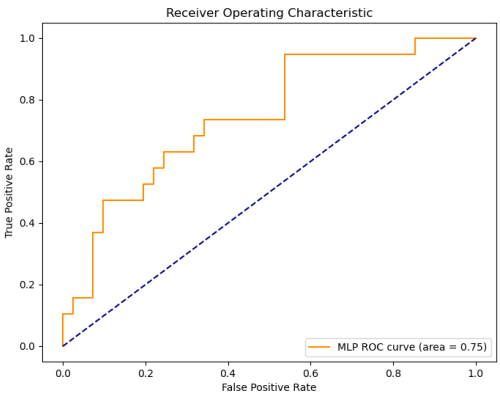

Figure 4: ROC curves comparing model performance.

Table 1: Model Performance Comparison

Model	Accuracy	AUC
Logistic Regression	0.817	0.859
MLP Classifier	0.750	0.746

3 Discussion

The superior performance of Logistic Regression suggests that simpler linear models are less prone to overfitting on limited clinical samples (N=299) compared to complex neural network architectures.

References

[1] Chicco, D., & Jurman, G. (2020). Machine learning can predict survival of patients with heart failure. *BMC Medical Informatics*, 20(1).